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Abstract

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Introduzione

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Parole chiave: qui, vanno, le parole chiave, della tesi

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1 | Introduction

2 | Fundamentals

2.1. Quantum computing

Quantum computing is a *new* computing paradigm based on the theory of quantum mechanics.

2.1.1. Threats to classical cryptography

2.1.2. Post Quantum Cryptography

2.2. Abstract Algebra

Definition 2.2.1 (Group)**.** Let G be a set. The structure $(G, \cdot)$ is a *group* if it satisfies the following axioms:

- Closure: For all $a, b \in G$, $a \cdot b \in G$.
- Associativity: For all $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- Identity: There exists an element $e \in G$ such that for all $a \in G$, $e \cdot a = a \cdot e = a$.
- Inverse: For all $a \in G$, there exists an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

If the group also satisfies the commutative property (i.e. $a \cdot b = b \cdot a$ for all $a, b \in G$), then it is called an *abelian group*.

Commonly, we call $(G, +)$ an *additive* group and $(G, \cdot)$ a *multiplicative* group. In this case, the identity element are denoted by 0 and 1, respectively.

Definition 2.2.2 (Field)**.** Let K be a set. The structure $(K, +, \cdot)$ is a *field* if it satisfies the following properties:

- Additive group: $(K, +)$ is an abelian group.
- Multiplicative group: $(K \setminus \{0\}, \cdot)$ is an abelian group.
- Distributivity: For all $a, b, c \in K$, $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

Definition 2.2.3 (Characteristic)**.** Let K be a field. The *characteristic* of K is the smallest positive integer n such that

$$\underbrace{1 + \cdots + 1}_{n \text{ times}} = 0$$

Definition 2.2.4 (Order)**.** Let K be a field. The *order* of K is the amount of elements in K; if the order of K is finite, it is also called a Galois field. A Galois field of order q, denoted as $\mathbb{F}_q$ or $GF(q)$, exists if and only if $q = p^n$ for some prime p; all Galois fields of order q are isomorphic.

Galois fields can be constructed in two ways:

- Primitive fields: $GF(p) \cong \mathbb{Z}_p$, with p prime, is a field of order and characteristic p.
- $GF(p^n) = GF(p)[X]/P(X)$ with $P(X)$ irreducible polynomial in $GF(p)[X]$ of degree n, is a field of order p^n and characteristic p.

Definition 2.2.5 (Field extension)**.** Let K and L be fields. If $K \subseteq L$, then we say that K is a *subfield* of L; conversely, we say that L is a *field extension* of K. Given a field extension, denoted as L/K, the field L is a vector space over K; the dimension of the vector space defines the *degree of the extension*, and is denoted by $[L : K]$. In the case of finite fields, $[GF(p^n) : GF(p)] = n$.

> **Example: Binary fields**
> The smallest Galois field is $GF(2)$, which is isomorphic to $\mathbb{Z}_2 = \{0, 1\}$. We can build $\mathbb{F}_8 = GF(2^3)$ as $GF(2)[X]/\langle X^3 + X + 1\rangle$. The elements of $\mathbb{F}_8$ are then the binary polynomials of degree strictly less than $n = 3$; to avoid confusion, the elements are written as polynomials in the variable α, where $P(\alpha) = 0$.
> We have that $\mathbb{F}_8 = \{0, 1, \alpha, 1 + \alpha, \alpha^2, 1 + \alpha^2, \alpha + \alpha^2, 1 + \alpha + \alpha^2\}$, so the order of the field is 8. Moreover, since $1 + 1 = 0$ in $GF(2)$, we have that $1 + 1 = 0$ in $\mathbb{F}_8$ as well, so the characteristic of $\mathbb{F}_8$ is 2.
> By construction, $\mathbb{F}_8$ is a field extension of $GF(2)$ of degree 3; the vector space $\mathbb{F}_8$ is thus a vector space over $GF(2)$ of dimension 3, having as basis the elements $\{1, \alpha, \alpha^2\}$.

2.3. Coding Theory

In this section we give the basic concepts and definitions of coding theory ([MS77]), required to formally define the problems on which HQC and similar cryptosystems are based.

2.3.1. General definitions

We denote with $\mathbb{F}_2$ the binary finite field and with $\mathbb{Z}$ the integer ring.
We define $\mathcal{V} = \mathbb{F}_2^n$ the vector space of dimension n over $\mathbb{F}_2$ and $\mathcal{R} = \mathbb{F}_2[X]/(X^n - 1)$ the ring of binary polynomials of degree less than n for some $n \in \mathbb{N}$.
Additionally, we denote by $\mathtt{wt}(\cdot)$ the Hamming weight of a vector *i.e.* the number of its non-zero coordinates.

Definition 2.3.1 (Circulant matrix)**.** Let $\mathbf{v} = (v_0, \cdots, v_{n-1}) \in \mathcal{V}$. The circulant matrix induced by $\mathbf{v}$ is defined as a matrix whose i-th row is the vector $\mathbf{v}$ shifted by i positions to the right:

$$\mathbf{circ}(\mathbf{v}) = \begin{pmatrix} v_0 & v_1 & \cdots & v_{n-1} \\ v_{n-1} & v_0 & \cdots & v_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ v_1 & v_2 & \cdots & v_0 \end{pmatrix} \in \mathbb{F}_2^{n \times n}$$

The algebra of $n \times n$ circulant matrices over $\mathbb{F}_2$ is isomorphic to the algebra of polynomials in the ring $\mathcal{R}$:

$$\mathbf{circ}(\mathbf{v}) \simeq v_0 + v_1 x + v_2 x^2 + \cdots + x_{n-1} x^{n-1}$$

Definition 2.3.2 (Linear q-ary Codes)**.** A Linear q-ary code $\mathcal{C}$ of lenght n and dimension k (denoted as $[n, k]$) is a subspace of $\mathbb{F}_q^n$.
Due to their vast use in cryptography (and in particular in this work), the following definitions and results are relative to binary codes ($q = 2$).

Definition 2.3.3 (Linear Codes)**.** A linear code $\mathcal{C}$ of lenght n and dimension k (denoted as $[n, k]$) is a subspace of $\mathcal{V}$ of dimension k; the elements of $\mathcal{C}$ are called *codewords*.
Due to the linearity of $\mathcal{V}$, any linear combination of two codewords is also a codeword.

Definition 2.3.4 (Generator Matrix)**.** Let $\mathcal{C} = [n, k]$ be a linear code. The matrix $\mathbf{G} \in \mathbb{F}_2^{k \times n}$ is called the generator matrix of $\mathcal{C}$ if

$$\mathcal{C} = \{\mathbf{mG}, \text{ for } \mathbf{m} \in \mathbb{F}_2^k\}$$

Definition 2.3.5 (Parity-Check Matrix)**.** Let $\mathcal{C} = [n, k]$ be a linear code. The matrix $\mathbf{H} \in \mathbb{F}_2^{(n-k) \times n}$ is called the parity-check matrix of $\mathcal{C}$ if

$$\mathcal{C} = \{\mathbf{v} \in \mathbb{F}_2^n \text{ such that } \mathbf{H}\mathbf{v}^\top = 0\}$$

In other words, $\mathbf{H}$ is the generator matrix of the dual code $\mathcal{C}^{\perp}$.

Definition 2.3.6 (Rank)**.** Let $\mathcal{C} = [n, k]$ be a linear code. The rank of $\mathcal{C}$ is defined as the dimension of the space spanned by its codewords or, in other words, the rank of its parity-check matrix.

$$r(\mathcal{C}) = r(\mathbf{H}) = n - k$$

A high rank code can correct more errors with fewer parity-check equations, but it usually is harder to construct and decode.

Definition 2.3.7 (Syndrome)**.** Let $\mathcal{C} = [n, k]$ be a code, $\mathbf{H}$ its parity matrix and $\mathbf{v} \in \mathbb{F}_2^n$ a word. The syndrome of $\mathbf{v}$ is defined as

$$\sigma(\mathbf{v}) = \mathbf{H}\mathbf{v}^{\top}$$

From this definition, we can say that a vector $\mathbf{v}$ is a codeword if and only if its syndrome is the zero vector.

Definition 2.3.8 (Rate)**.** Let $\mathcal{C} = [n, k]$ be a code. The rate of $\mathcal{C}$ is defined as $R = \frac{k}{n}$.

The rate of a code is a measure of its efficiency: with a higher rate we can transmit more information with the same amount of bits, at the cost of reduced redundancy (and thus reduced error correcting capability).

Definition 2.3.9 (Minimum distance)**.** Let $\mathcal{C} = [n, k]$ be a code and ω a norm on $\mathcal{V}$. The minimum distance of $\mathcal{C}$ is defined as

$$d = \min_{\mathbf{v},\mathbf{w} \in \mathcal{C}, \mathbf{v} \neq \mathbf{w}} \omega(\mathbf{v} - \mathbf{w})$$

The length, dimension and minimum distance of a linear code $\mathcal{C} = [n, k, d]$ are tied by the Singleton bound [Sin64]:

$$d \leq n - k + 1$$

If the equality holds, $\mathcal{C}$ is called a maximum distance separable (MDS) code, and it has desirable properties in terms of error correction.

Definition 2.3.10 (Error Correction Capability)**.** A linear code $\mathcal{C} = [n, k, d]$ can decode arbitrary codewords with at most $\delta = \lfloor \frac{d-1}{2} \rfloor$ errors. If $\mathcal{C}$ is a MDS code, we can rewrite this as $\delta = \lfloor \frac{n-k}{2} \rfloor$.

Definition 2.3.11 (Gilbert-Varshamov bound)**.** Let $\mathcal{C} = [n, k]$ be a linear code. The Gilbert-Varshamov bound [Gil52, Var57] states that $\mathcal{C}$ can correct at least t errors if its

size satisfies

$$2^k \geq \sum_{i=0}^{t} \binom{n}{i} \tag{2.1}$$

2.3.2. Decoding methods

https://en.wikipedia.org/wiki/Decoding_methods

Minimum distance decoding

Maximum likelihood decoding

Syndrome decoding

2.3.3. Code families

Informally, when we say that a code is *good*, we mean that it satisfies some properties, such as:

- High minimum distance: the higher the minimum distance, the more errors the code can correct.
- High rate: the higher the rate, the more information we can transmit with the same amount of bits.
- Efficiency: we want the encoder and decoder to be fast and simple to implement.

In this section we will describe some of the most common code families and the properties they satisfy.

Definition 2.3.12 (Cyclic Codes)**.** Let $\mathbf{c} = (c_0, \cdots, c_{n-1}) \in \mathcal{V}$ be a vector. A linear code $\mathcal{C} = [n, k]$ is Cyclic if, for any $\mathbf{v} \in \mathcal{C}$, the vector obtained by applying a circular shift to $\mathbf{v}$ is also a codeword. codeword. If we think of $\mathbf{v}$ as a polynomial in $\mathcal{R}$, we can say that $\mathcal{C}$ is cyclic if, for any $\mathbf{v} \in \mathcal{C}$, the polynomial $X \cdot \mathbf{v}$ is also a codeword.

Definition 2.3.13 (Quasi-Cyclic Codes)**.** Let $\mathbf{c} = (\mathbf{c}_0, \cdots, \mathbf{c}_{s-1}) \in \mathbb{F}_2^{sn}$ be a vector, where we consider each $\mathbf{c}_i$ as a vector in $\mathbb{F}_2^n$. A linear code $\mathcal{C} = [sn, k]$ is Quasi-Cyclic of index s if, for any $\mathbf{c} \in \mathcal{C}$, the vector obtained by applying a circular shift to every block $\mathbf{c}_i$ simultaneously is also a codeword. QC codes are a generalization of cyclic codes.

Definition 2.3.14 (Double Circulant Codes)**.** A code $\mathcal{C}$ is double circulant if it is Quasi-Cyclic of even length and index $s = 2$.

Random codes

In a linear random code the codewords are chosen at random; specifically, we can build the parity matrix $\mathbf{H}$ of the code by randomly sampling each entry from a Bernoulli distribution.
For sufficiently large n, random linear codes reach the Gilbert-Varshamov bound; moreover, the probability of decoding error decreases exponentially as the minimum distance increases. On the other hand, the computational complexity of a decoder for a random code grows exponentially with the length of the code.

Repetition codes

Repetition codes are among the simplest codes, as they are just a repetition of the same vector k times. Repetition codes can be easily decoded by majority voting, making the decoding process very fast. However, they have a very low rate and they are not efficient in terms of error correction capability.

Concatenated codes

Concatenated codes [For65] are built by combining an *inner* code $\mathcal{C}_{in}$ and an *outer* code $\mathcal{C}_{out}$. The goal of concatenated codes is to obtain a code which has both a polynomial-time decoder and exponentially decreasing error probability with increased block length.

Reed-Muller codes

Reed-Muller codes [Ree53, Mul54] are a family of codes; they are usually denoted as $\mathsf{RM}(r, m)$, as they have a message length which is equal to

$$k = \sum_{i=0}^{r} \binom{m}{i}$$

RM codes are:

- Locally testable: it is possible to determine whether a string is a word in the code by testing only a few bits of the string.
- Locally decodable: it is possible to recover the original information by using a small amount of bits of the codeword.
- List decodable: TODO

Due to their properties, we can describe RM codes based on low degree polynomials.

2.3.4. Difficult problems for cryptography

The problems we are interested in are all variants of the *decoding problem*, which requires to find the closest codeword to a given word with respect to a certain metric (the Hamming weight in our case).
In the case of linear codes we have shown that the decoding problem stays the same whether one is given the syndrome of the received vector or the vector itself; for this reason, we will only consider the *Syndrome Decoding* (SD) problem.

Definition 2.3.15 (SD Distribution)**.** Given $n, k, w \in \mathbb{N}$, the distribution $\mathsf{SD}(n, k, w)$ randomly samples a parity-check matrix $\mathbf{H} \xleftarrow{\$} \mathbb{F}_2^{(n-k)\times n}$, a vector $\mathbf{x} \xleftarrow{\$} \mathbb{F}_2^n$ such that $\mathtt{wt}(\mathbf{x}) = w$ and returns $(\mathbf{H}, \sigma(\mathbf{x}))$.

Definition 2.3.16 (SD Problem)**.** Given $(\mathbf{H}, \mathbf{y}^\top) \in \mathbb{F}_2^{(n-k)\times n} \times \mathbb{F}_2^{(n-k)}$ from the $\mathsf{SD}(n, k, w)$ distribution, the SD problem requires to find a vector $\mathbf{x} \in \mathbb{F}_2^n$ such that $\sigma(\mathbf{x}) = \mathbf{y}^\top$ and $\mathtt{wt}(\mathbf{x}) = w$.

This problem has been proven to be NP-complete in [BMVT78].

Definition 2.3.17 (Decisional SD Problem)**.** Given $(\mathbf{H}, \mathbf{y}^\top) \in \mathbb{F}_2^{(n-k)\times n} \times \mathbb{F}_2^{(n-k)}$, the DSD problem requires to determine whether $(\mathbf{H}, \mathbf{y}^\top)$ comes from the SD distribution or from the random distribution over $\mathbb{F}_2^{(n-k)\times n} \times \mathbb{F}_2^{(n-k)}$.

3 | Side Channel Attacks

Many cryptographic algorithms are considered unbreakable, regardless of the amount of computing power available. These secure cryptosystems however do not run in a vacuum with spherical cows, and have to be implemented, either in hardware or in software.
Regardless of the means of the implementation, the electronic devices that run these cryptosystems often *leak* information about the internal state of the device, potentially exposing secrets. But how does a device leak information?

A *side channel* is some observable aspect of a cryptosystem that reveals information held within that system [Sta10]; example of side channels can be the execution time of an algorithm, the power consumption of a device or the electromagnetic radiation emitted while running some operations.
In this chapter we will describe the most common side channel attacks, how they can be used to retrieve secret information and how to mitigate them.

3.1. Timing attacks

In a timing attack, the attacker tries to infer information about a secret by measuring the time it takes to execute some operation: the idea behind this attack is that the time it takes to execute an operation depends on the input of the operation, and thus by measuring the execution time of the operation we can infer information about the input.

To better explain how timing attacks work, let us consider the following code snippet:

```
1  int check_pin(int *pin) {
2      int correct_pin[4] = {3, 1, 4, 2};
3
4      for (int i = 0; i < 4; i++) {
5          if (pin[i] != correct_pin[i]) {
6              return 0;
7          }
8      }
9      return 1;
10 }
```

We can think this function as a part of a bigger system, where the user enters a PIN and the system checks if it is correct. The function simply checks if the PIN entered by the user is equal to the correct PIN, and returns 1 if it is correct and 0 otherwise. Being the PIN composed of decimal digits, the total amount of combinations is 10^4, so on average we would need 5000 guesses to find the correct PIN; however, we can reduce the number of guesses needed to find the correct PIN by using a timing attack.

We can observe that the function returns as soon as a digit of the PIN is incorrect: under the reasonable assumption that we can measure the execution time of the function, we can guess the correct PIN one digit at a time. Starting from 0000, we cycle through all the possibilities for the first digit $(0000, 1000, \cdots, 9000)$ and measure the execution time: out of the 10 measurements, the one with the highest execution time corresponds to the correct digit; we can then repeat the same process for the remaining digits, so that after at most 40 guesses we find the correct pin.

Even though this attack strategy may seem trivial and easy to mitigate, it can still wreck havoc: real-life examples of timing attacks include, for example, Spectre [KHF+19] and Meltdown [LSG+18], which affected almost all modern processors and allowed an attacker to read the memory of any process running on the same machine.

3.2. Power attacks

4 | Hamming Quasi-Cyclic

Hamming Quasi-Cyclic [MAB+18] is a fourth round candidate in the NIST Post-Quantum cryptography standardization process. HQC is a code-based cryptosystem, that includes comprises both a public-key encryption (HQC.PKE) and a key encapsulation mechanism (HQC.KEM).
In this section we will describe the inner workings of HQC and show its theoretical security; moreover, we will give an overview of the currently known attacks against HQC and their practical impact.

4.1. Encryption scheme

We can think of a generic encryption scheme as a tuple of four polynomial time algorithms:

- $\mathsf{Setup}(1^\lambda)$: generates the global parameters param based on the security parameter λ.
- $\mathsf{Keygen}(\mathsf{param})$: generates a public key pk and a secret key sk based on the global parameters param.
- $\mathsf{Encrypt}(\mathsf{pk}, \mathbf{m}, \theta)$: outputs the ciphertext $\mathbf{c}$ for the message $\mathbf{m}$ with the randomness θ under the public key pk.
- $\mathsf{Decrypt}(\mathsf{sk}, \mathbf{c})$: outputs the message $\mathbf{m}$ for the ciphertext $\mathbf{c}$ (or $\perp$ if the ciphertext is invalid).

TODO: add security proof?

4.2. Public Key Encryption

HQC uses two different codes:

- A publicly available decodable code $\mathcal{C} = [n, k]$ generated by $\mathbf{G} \in \mathbb{F}_2^{k \times n}$ with an efficient decoding algorithm $\mathcal{C}.\mathsf{Decode}()$ which can correct up to δ errors.

- A random double-circulant $[2n, n]$ code with parity-check matrix $\mathbf{H} = (\mathbf{I_n}, \mathbf{circ}(\mathbf{h}))$.

The HQC.PKE defines the four algorithms of an encryption scheme as follows:

- Setup(1^λ): generates the set of parameters $\mathsf{param} = (n, k, \delta, \omega, \omega_r, \omega_e)$
- Keygen(param): samples $\mathbf{h} \xleftarrow{\$} \mathcal{R}$ and $(\mathbf{x}, \mathbf{y}) \xleftarrow{\$} \mathcal{R} \times \mathcal{R}$ such that $\mathtt{wt}(\mathbf{x}) = \mathtt{wt}(\mathbf{y}) = \omega$; the keypair is defined as $\mathsf{sk} = (\mathbf{x}, \mathbf{y})$ and $\mathsf{pk} = (\mathbf{h}, \mathbf{s} = \mathbf{x} + \mathbf{h} \cdot \mathbf{y})$.
- Encrypt(pk, $\mathbf{m}$, θ): samples $\mathbf{e} \xleftarrow{\$} \mathcal{R}$, $(\mathbf{r_1}, \mathbf{r_2}) \xleftarrow{\$} \mathcal{R} \times \mathcal{R}$ such that $\mathtt{wt}(\mathbf{e}) = \omega_e$ and $\mathtt{wt}(\mathbf{r_1}) = \mathtt{wt}(\mathbf{r_2}) = \omega_e$; the ciphertext is defined as the pair $(\mathbf{u}, \mathbf{v})$, where $\mathbf{u} = \mathbf{r_1} + \mathbf{h} \cdot \mathbf{r_2}$ and $\mathbf{v} = \mathbf{mG} + \mathbf{s} \cdot \mathbf{r_2} + \mathbf{e}$.
- Decrypt(sk, $\mathbf{c}$): outputs $\mathcal{C}$.Decode$(\mathbf{v} - \mathbf{u} \cdot \mathbf{y})$.

The idea at the core of HQC.PKE is to encode the message ($\mathbf{mG}$) and then add random errors that can be removed only by knowing the secret key.

TODO: add security proof?

4.3. Key Encapsulation Mechanism

In practice, public key encryption is not used to encrypt large amounts of data because of its high exectution time compared to symmetric encryption. For this reason, we use public key encryption to exchange a shared secret key, which is then used to encrypt the actual data: the practice of using a public key encryption scheme to exchange a shared secret key is referred to as key encapsulation mechanism (KEM).
The Fujisaki-Okamoto transformation [FO13] is a technique that allows to transform a *weak* IND-CPA encryption scheme into a *stronger* IND-CCA2 key encapsulation mechanism.

Let $\mathcal{E}$ an instance of HQC.PKE and $\mathcal{G}, \mathcal{H}$ and $\mathcal{K}$ hash functions. The HQC.KEM scheme is defined as follows:

- Setup(1^λ): generates the set of parameters as in HQC.PKE.Setup, where k will be the key size.
- Keygen(param): generates a keypair as in HQC.PKE.Keygen.
- Encapsulate(pk): sample $\mathbf{m} \xleftarrow{\$} \mathbb{F}_2^k$ to derive the randomness $\theta = \mathcal{G}(\mathbf{m})$. Generate the ciphertext as $\mathbf{c} = (\mathbf{u}, \mathbf{v}) \leftarrow \mathcal{E}.\mathsf{Encrypt}(\mathsf{pk}, \mathbf{m}, \theta)$ and derive the symmetric key as $K \leftarrow \mathcal{K}(\mathbf{m}, \mathbf{c})$; send $(\mathbf{c}, \mathbf{d} = \mathcal{H}(\mathbf{m}))$.

- $\mathsf{Decapsulate}(\mathsf{sk}, \mathbf{c}, \mathbf{d})$: decrypt the ciphertext as $\mathbf{m}' = \mathcal{E}.\mathsf{Decrypt}(\mathsf{sk}, \mathbf{c})$, compute $\theta' = \mathcal{G}(\mathbf{m}')$ and re-encrypt to get $\mathbf{c}' \leftarrow \mathcal{E}.\mathsf{Encrypt}(\mathsf{pk}, \mathbf{m}', \theta)$. If $\mathbf{c} \neq \mathbf{c}'$ or $\mathbf{d} \neq \mathcal{H}(\mathbf{m}')$ return $\perp$, otherwise derive the shared key $\mathcal{K}(\mathbf{m}', \mathbf{c}')$.

TODO: what happens if we do not check $\mathbf{d} == \mathcal{H}(\mathbf{m}')$?

5 | Multiplication Algorithms

In this chapter etc etc

5.1. Multiplication 101

5.1.1. Schoolbook multiplication

5.1.2. Integers and polynomials

Multiply by value

If we think of the n digits of each input as coefficients of a polynomial, we can calculate the product by following these steps:

1. Sample the polynomials at $2n$ points
2. Perform the pointwise multiplication of the evaluations
3. Interpolate the result to obtain the coefficients of the product

Let $a = 123$ and $b = 45$ be two integers. We can calculate their product $c = 123 \cdot 45$

$$a = 123 \longrightarrow a(x) = 1x^2 + 2x + 3$$
$$b = 45 \longrightarrow b(x) = 0x^2 + 4x + 5$$

We sample the polynomials at $2n = 6$ points:

$$\begin{aligned} a(0) &= 1 \cdot 0^2 + 2 \cdot 0 + 3 = 3 & b(0) &= 0 \cdot 0^2 + 4 \cdot 0 + 5 = 5 \\ a(1) &= 1 \cdot 1^2 + 2 \cdot 1 + 3 = 6 & b(1) &= 0 \cdot 1^2 + 4 \cdot 1 + 5 = 9 \\ a(2) &= 1 \cdot 2^2 + 2 \cdot 2 + 3 = 11 & b(2) &= 0 \cdot 2^2 + 4 \cdot 2 + 5 = 13 \\ a(3) &= 1 \cdot 3^2 + 2 \cdot 3 + 3 = 18 & b(3) &= 0 \cdot 3^2 + 4 \cdot 3 + 5 = 17 \\ a(4) &= 1 \cdot 4^2 + 2 \cdot 4 + 3 = 27 & b(4) &= 0 \cdot 4^2 + 4 \cdot 4 + 5 = 21 \\ a(5) &= 1 \cdot 5^2 + 2 \cdot 5 + 3 = 38 & b(5) &= 0 \cdot 5^2 + 4 \cdot 5 + 5 = 25 \end{aligned}$$

The pointwise multiplication of the evaluations is:

$$\begin{aligned} c(0) &= a(0) \cdot b(0) = 3 \cdot 5 = 15 \\ c(1) &= a(1) \cdot b(1) = 6 \cdot 9 = 54 \\ c(2) &= a(2) \cdot b(2) = 11 \cdot 13 = 143 \\ c(3) &= a(3) \cdot b(3) = 18 \cdot 17 = 306 \\ c(4) &= a(4) \cdot b(4) = 27 \cdot 21 = 567 \\ c(5) &= a(5) \cdot b(5) = 38 \cdot 25 = 950 \end{aligned}$$

Finally, we can interpolate the result to obtain the coefficients of the product:

$$c(x) = 4x^3 + 13x^2 + 22x + 15$$

and convert back to an integer by shifting and adding the coefficients, finding that $123 \cdot 45 = 5535$.

This approach performs the pointwise multiplication in $\mathcal{O}(n)$, but requires $\mathcal{O}(n^2)$ operations to perform the sampling step and $\mathcal{O}(n^3)$ operations to interpolate the result, making it worse than the classical method.

5.1.3. Faster multiplication algorithms

5.2. FFT-Based multiplication

5.2.1. Convolutions and Fourier Transforms

We are however free to choose the points at which we sample the polynomials, which can drastically reduce the amount of operations needed. Instead of sampling at $2n$ random points, we can choose the points to be the N-th roots of unity, *i.e.* we calculate the Discrete Fourier Transform of the input polynomials:

$$\mathcal{F}(x) = \sum_{n=0}^{N} x_n e^{-\frac{i2\pi}{N}kn} \tag{5.1}$$

At this point we can perform the pointwise product in the *transformed* domain and, since $\mathcal{F}$ is invertible, apply the inverse transform:

$$\mathcal{F}^{-1}(X) = \frac{1}{N}\sum_{n=0}^{N} X_n e^{i\frac{2\pi}{N}kn} \tag{5.2}$$

The algorithms that compute the discrete Fourier Transform of a sequence (or its inverse) are called fast Fourier transforms (FFT), and have an asymptotic complexity of $\mathcal{O}(n \log n)$.

5.2.2. Fast Fourier Transform

Cooley-Tukey FFT

The Cooley-Tukey FFT [CT65] is the most common FFT algorithm. It is a *divide et impera* algorithm, which recursively expresses a DFT of size $N = N_1 N_2$ into N_1 smaller DFTs of size N_2 each. The simplest form of the Cooley-Tukey algorithm is the radix-2 decimation-in-time FFT, which computes the DFTs of the even and odd indexed inputs separately and then combines the results. Assuming that N is a power of 2, we can rewrite the DFT as:

$$\begin{aligned}
X_k &= \sum_{n=0}^{N-1} x_n e^{-\frac{2\pi i}{N}nk} \\
&= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N}(2m)k} + \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N}(2m+1)k} \\
&= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N/2}mk} + e^{-\frac{2\pi i}{N}k} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N/2}mk} \\
&= E_k + e^{-\frac{2\pi i}{N}k} O_k \qquad \text{for } k = 0, \cdots, N/2-1
\end{aligned}$$

where we split the DFT into two smaller DFTs, on the even and odd indexes respectively. Because of the periodicity of the complex exponential, we can also write:

$$\begin{aligned}
X_{k+N/2} &= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N/2} m(k+N/2)} + e^{-\frac{2\pi i}{N}(k+N/2)} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N/2} m(k+N/2)} \\
&= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N/2} mk} e^{-2\pi mi} + e^{-\frac{2\pi i}{N} k} e^{-\pi i} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N/2} mk} e^{-2\pi mi} \\
&= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N/2} mk} - e^{-\frac{2\pi i}{N} k} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N/2} mk} \\
&= E_k - e^{-\frac{2\pi i}{N} k} O_k \qquad \text{for } k = 0, \cdots, N/2-1
\end{aligned}$$

This means that we can express a DFT of size N as two DFTs of size $N/2$, where we can re-use intermediate results to calculate multiple outputs. Since the base case of the recursion has complexity $\mathcal{O}(n)$ we get the recurrence relation $T(n) = 2T(n/2) + \mathcal{O}(n)$; following the Master theorem [BHS80], the computational complexity of the Cooley-Tukey FFT is $\mathcal{O}(n \log n)$.

Gao-Mateer FFT

The Cooley-Tukey FFT works when the n points to sample are n-th roots of unity, and hence form a cyclic multiplicative group; this and similar algorithms are also referred to as *multiplicative* FFTs, as they are based on the factorization of $x^n - 1$.
Unfortunately, multiplicative FFTs do not work when the underlying field does not admit primitive roots of unity, as it happens for fields of characteristic two; for this reason, *additive* FFTs were invented [WZ88, Can89]. Instead of using the n-th roots of unity, additive FFTs evaluate the polynomials at the powers of a primitive $(n-1)$-th root of unity, as well as the zero element.
The Gao-Mateer FFT [GM10] is an algorithm that computes the additive FFT over finite fields of characteristic two.

Cantor basis In [Can89] Cantor showed how to get a special basis in which it is possible to compute the FFT efficiently for any $m = p^n$, with p a prime number; later in [GM10] Gao and Mateer used a simpler construction for the special case $p = 2$, where the Cantor basis $\mathbf{v} = (v_0, \cdots, v_{m-1})$ follows the structure:

$$v_0 = 1, \quad v_i^2 + v_i = v_{i-1} \text{ for } i \geq 1 \tag{5.3}$$

With respect to the Cantor basis $\mathbf{v}$, we define its sequence of subspaces as:

$$W_0 := \{0\}, \quad W_i := span\{v_0, \cdots, v_{i-1}\} \text{ for } i \geq 1 \tag{5.4}$$

By construction, we have that $W_0 \subset W_1 \subset \cdots \subset W_{m-1} = \mathbb{F}_{2^m}$ and the basis for a smaller space is contained into the basis of a larger space: this means that we can arbitrarily treat elements of smaller fields as elements in the larger fields by padding zeros to the extra dimensions.

With respect to each element of the basis v_i we can define its subspace polynomial $s_i(x)$, also called vanishing polynomial, as:

$$s_i(x) = \prod_{a \in W_i} (x - a) \tag{5.5}$$

Since $\dim(W_i) = i$ we have $\deg(s_i(x)) = |W_i| = 2^i$. Such polynomials, as shown in [Can89, GM10], satisfy the following useful properties:

- linearity: $s_i(x + y) = s_i(x) + s_i(y)$
- recursivity: $s_{i+j}(x) = s_i(s_j(x))$

Divide et Impera As with the other FFTs, the result of the Gao-Mateer FFT is a set of evaluation points of a polynomial. Let $f(x) \in \mathbb{F}_{2^m}[x]$ be a polynomial of degree $d \leqslant n$ with $n = 2^\ell, \ell \leqslant m$ and $v \in \mathbb{F}_{2^m}$. The result of the AFFT is the set of evaluation points:

$$S = \{f(\alpha) | \alpha \in v + W_\ell\} \tag{5.6}$$

For $\ell \geqslant 0$, the first step of the AFFT performs a *radix conversion*, transforming the input polynomial $f(x)$ into:

$$f(x) = f^{(0)}(s_1(x)) + x \cdot f^{(1)}(s_1(x)) \tag{5.7}$$

where $f^{(0)}$ and $f^{(1)}$ are polynomials of degree at most $2^{\ell-1}$ and can be calculated by using only field additions.
At this point, if we have $S^{(j)} = \{f^{(j)}(s_1(\alpha)) | \alpha \in v + W_\ell\}$ for $j \in \{0, 1\}$, then we can compute each $f(\alpha)$ as $f^{(0)}(\alpha) + \alpha \cdot f^{(1)}(\alpha)$. Because of the linearity of the vanishing polynomial, we can also observe that $s_1(v + W_\ell) = s_1(v) + W_{\ell-1}$, and thus compute $S^{(0)}$ and $S^{(1)}$ recursively, reducing a problem of size n to two problems of size $n/2$.

Segmentation of large polynomials

Suppose we are multiplying two binary polynomials $f(x)$ and $g(x) \in \mathbb{F}_2[x]$, represented as sequences of d bits:

$$a(x) = a_0 + a_1 x + \cdots + a_{d-1} x^{d-1}$$

In some cases, especially with high degree polynomials, it can be useful to split the polynomials into smaller segments, which can be easier to handle. The standard *Kronecker segmentation* for multiplying binary polynomials is performed as follows

1. Split the polynomials into $n = \lceil d/w \rceil$ w-bit blocks:

$$\begin{aligned} a(x) &= a_0 + a_1 x + \cdots + a_{d-1} x^{d-1} \\ &= (a_0 + \cdots + a_{w-1} x^{w-1}) + \cdots + (a_{d-w} + \cdots + a_{d-1} x^{w-1}) x^{w(n-1)} \end{aligned}$$

2. Define the field $\mathbb{F}_{2^{2w}} := \mathbb{F}_2[z]/g(z)$, with $g(z)$ irreducible polynomial of degree $2w$.

3. Define the map $\psi : \mathbb{F}_2[x] \to \mathbb{F}_{2^{2w}[y]}$ such that:

$$a'(y) = \psi(a(x)) := a'_0 + a'_1 y + \cdots + a'_{n-1} y^{n-1}$$

   where $a'_i = (a_{iw} + a_{iw+1} z + \cdots + a_{iw+(w-1)} z^{w-1})$.

4. Calculate the product $c'(y) = a'(y) \cdot b'(y) = c'_0 + c'_1 y + \cdots \in \mathbb{F}_{2^{2w}}[y]$

5. Map $z \mapsto x$ and $y \mapsto x^w$, collect terms and coefficients and find the result.

At step (4) we employ the FFT-based method to perform the pointwise multiplication.

6 | Empirical Evaluations

7 | Conclusions

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A | Appendix A

Appendix

List of Figures

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Acknowledgements

Here you might want to acknowledge someone.